Supervised Learning In Polyp Segmentation in Medical Image Analysis

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Abstract. In recent years, convolutional neural network architectures have significantly advanced the field of computer vision, particularly in biomedical image segmentation tasks. Polyp segmentation technology based on computer vision plays an important role in diagnosing colorectal cancer by providing high segmentation accuracy. The development of specialized architectures such as U-Net, U-Net++, and ResU-Net has led to remarkable progress in improving segmentation accuracy in medical image analysis. In this report, we aim to explore the effectiveness of attention mechanisms in the Residual U-Net in combination with the lightweight MobileNetV2 backbone, called AGResU-NetV2. AGResU-NetV2 aims to solve the challenge of substantial computational resources and limiting deployment on low-power devices like IoT devices.

Keywords: Supervised Learning · Polyp Segmentation · Medical Image Analysysis · U-Net · MobileNetV2

1 Introduction

1.1 Background

Semantic Segmentation is a fundamental task in Computer Vision, where the input images are processed to generate masks highlighting regions of interest. This technique plays a crucial role in detecting the location, shape, and size of objects within images. Its versatility finds widespread applications across diverse industries. In healthcare, semantic segmentation aids in detecting tumors and cancers, facilitating early diagnosis and treatment. Similarly, in the automotive sector, it enables the identification of vehicles, pedestrians, and road signs, contributing to the development of advanced driver assistance systems (ADAS) and autonomous vehicles.

In recent years, with continuous advancements in artificial intelligence (AI), especially the use of deep learning and convolutional neural networks (CNNs) that can automatically extract features in medical images and videos, companies are racing to create faster, smaller, and more efficient medical devices with the implementation of AI to improve doctor's daily work.

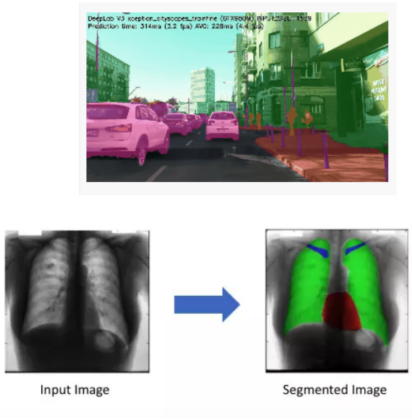


Fig. 1. Semantic segmentation

1.2 Motivation

Colorectal cancer (CRC) stands as one of the leading causes of cancer-related deaths worldwide. The significance of early polyp detection cannot be overstated, as it enables timely treatment for CRC. However, achieving highly accurate polyp segmentation still remains a formidable challenge for several reasons. First, there is a limited number of experts in the field. Doctors with extensive experience and expertise in colorectal cancer are scattered worldwide. Second, it is a labor-intensive and time-consuming process for healthcare professionals to label polyps. Furthermore, diagnosing pictures and videos carefully takes a significant amount of time. Due to safety concerns, endoscopic polypectomy can only be performed within a short timeframe. It requires doctors to be highly trained and experienced to detect polyps in a timely manner. Working in a highly stressful environment where the safety of human life is crucial, doctors are prone to fatigue and distraction, which can cause inconsistencies in diagnosis.

An effective strategy for addressing these obstacles involves advancing automated detection methods using AI. The implementation of AI can assist doctors in the early detection of colorectal cancer. By reducing human error, AI can automate the segmentation process and improve workflow by minimizing the likelihood of misdiagnosis and reducing the time to process images. This could lead to faster diagnosis and treatment for patients.

2 Related Work

There have been several methods developed, specifically for biomedical image segmentation, utilizing Convolutional Neural Networks for supervised learning. These methods often leverage the Encoder-Decoder architecture. [1]. The Encoder extracts relevant features from the images, while the Decoder takes these extracted features and reconstructs a segmentation mask.

2.1 U-Net

U-Net is a Convolutional Neural Network developed specifically for biomedical image segmentation. Its name derives from the U-shaped architecture it adopts, which consists of an encoder-decoder scheme. The encoder on the left reduces spatial dimensions while increasing the number of channels, effectively capturing the input's characteristics. The decoder upsamples the feature maps, restoring spatial dimensions while reducing channels. This design facilitates precise pixel-level predictions for the input image. Additionally, U-Net incorporates skip connections, allowing information from early layers to be directly passed to later layers. This mechanism preserves fine-grained details during the upsampling process. As a result, U-Net has demonstrated remarkable effectiveness in tasks such as cell segmentation and tissue classification, showcasing its versatility and robust performance in biomedical image analysis and beyond.[2]

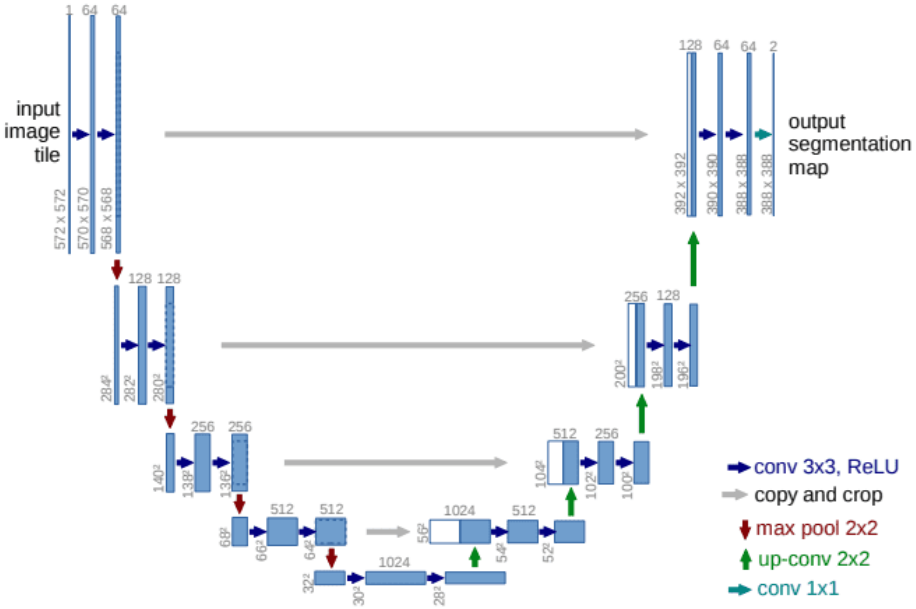


Fig. 2. U-Net Architecture

2.2 U-Net ++

U-Net ++ [3] represents an enhancement of the original U-Net, specifically tailored to improve performance in biomedical segmentation tasks. By refining the skip connections utilized in the original U-Net, U-Net++ accelerates the flow of feature maps from the encoder to the decoder, effectively integrating dissimilar

feature maps. The incorporation of skip pathways, in green and blue, ensures gradient flows throughout the network, thereby enhancing information propagation efficiency. Additionally, the implementation of deep supervision, highlighted in red, enables model pruning, thereby improving computational efficiency. With the aim of developing more efficient models for biomedical devices, U-Net++ represents a significant step forward in addressing the computational intensity challenge, making it feasible for implementation in smaller medical devices.

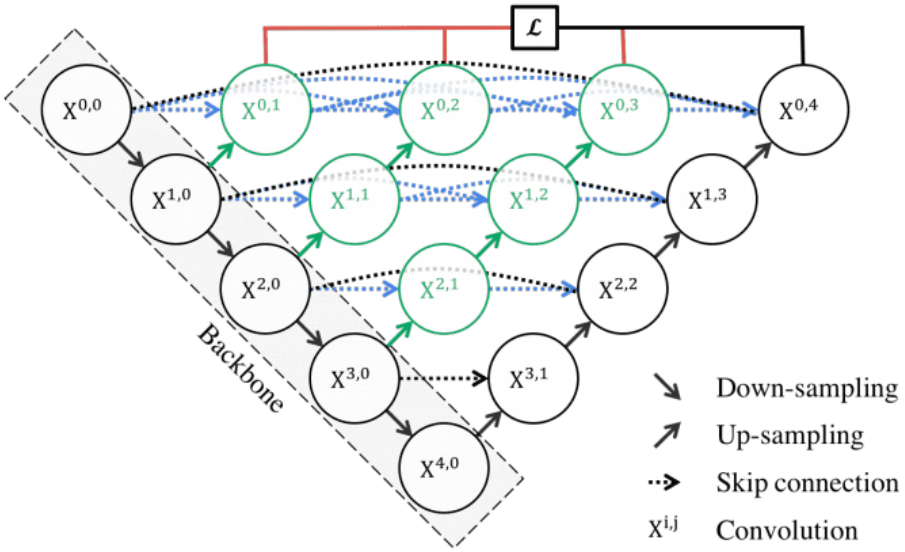


Fig. 3. U-Net++ Architecture

2.3 ResU-Net

ResU-Net refers to Deep Residual U-Net. It is an encoder-decoder architecture developed by Zhengxin Zhang et al. [4] for semantic segmentation. Originally designed for road extraction from high-resolution aerial images, ResUNet has since been adapted by researchers for tasks such as polyp segmentation, brain tumor segmentation, and more. ResU-Net combines the strengths of both U-Net and Deep Residual Learning. The integration of residual blocks allows for a deeper network architecture without encountering issues like vanishing or exploding gradients, leading to smoother training. Additionally, the implementation of rich skip connections enhances the flow of information between different layers of the network. These features collectively contribute to the model's robustness and effectiveness in various semantic segmentation tasks.

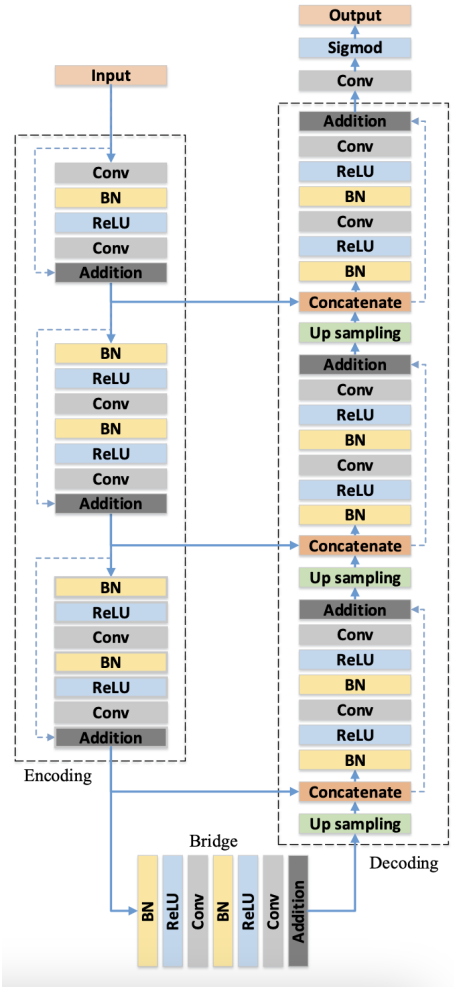


Fig. 4. ResU-Net Architecture

3 Indetified Methodologies

3.1 Problem Statement

The original U-Net architecture, along with its modifications such as U-Net++ and ResU-Net, has found wide application in various medical image segmentation tasks. However, despite their widespread use, these models require substantial computational resources and prolonged training times due to their complexity. Additionally, their resource-intensive nature makes them impractical for deployment on low-power devices like IoT devices. Consequently, while these models

demonstrate remarkable performance in tasks like polyp segmentation and other medical segmentation tasks, implementing them on IoT devices remains a challenge.

To address this challenge, there is a need for lightweight model architectures and optimization techniques. This involves two main focuses: first, reducing model complexity by focusing on relevant features of polyps, and second, reducing model computational resources by implementing lightweight, lower-computation models. By considering these aspects, a lightweight computing power model has been proposed, with details covered in the following section.

3.2 The Proposed Methods

I propose an ensemble method called AGResU-NetV2, which utilizes the Attention Gate ResU-Net [6] with a MobileNetV2 [7] backbone. This model combines the effectiveness of attention gates in the Residual U-Net architecture with the lightweight design of MobileNetV2, tailored for mobile and embedded vision applications. AGResU-NetV2 aims to tackle the challenge of implementing lightweight, high-accuracy models on low-power devices such as IoT devices.

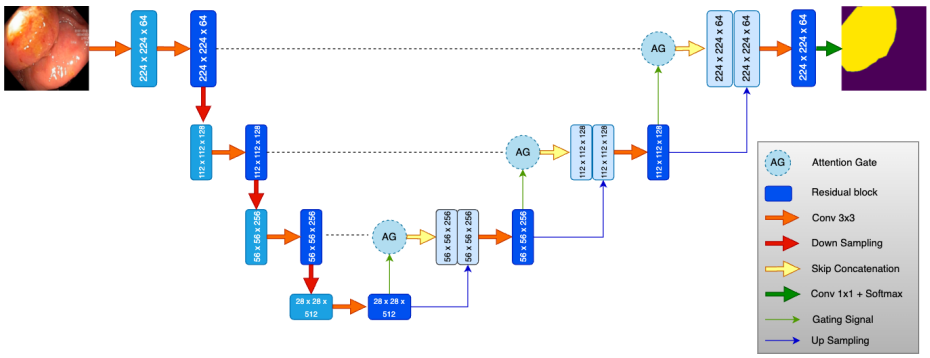


Fig. 5. AGResU-NetV2

The core AGRresU-Net shows a significant performance in MRI brain tumor segmentation and has potential in other medical segmentation task. According to Jianxin et al [6], AGResU-Net integrates residual modules and attention gates [8] into the original U-Net framework. By incorporating attention gate units into the skip connections, AGResU-Net focuses on important feature information while suppressing irrelevant and noisy features. This approach enriches feature learning by capturing intricate semantic details, particularly addressing the subtleties of small-scale brain tumors and similarly polyps tumors. The attention mechanisms embedded in AGResU-Net aim to enhance segmentation accuracy, especially in tasks such as brain tumor detection and other segmentation tasks.

MobileNetV2 is a convolutional neural network architecture designed for mobile and embedded vision applications, introduced by Google researchers in

2018.[7]. It builds upon the original MobileNet architecture with the aim of enhancing both efficiency and performance. One of the key innovations of MobileNetV2 is the utilization of depthwise separable convolutions. This technique divides the convolution operation into separate depthwise and pointwise convolutional layers, effectively reducing computational complexity while preserving representational capacity. Furthermore, MobileNetV2 introduces inverted residual blocks featuring linear bottleneck layers and shortcut connections. These architectural elements enable the construction of deeper networks with lower computational cost. As a result, MobileNetV2 is particularly well-suited for tasks such as semantic segmentation on resource-constrained devices like smartphones and IoT devices.

4 Conclusion and Future Work

4.1 Conclusion

This report addresses the use of supervised learning in polyp segmentation in medical image analysis.

In the first part of the report, three architectures of supervised learning using Convolutional Neural Networks in biomedical image segmentation are discussed, including U-Net, U-Net++, and ResU-Net. These architectures employ the Encoder-Decoder framework to extract relevant features from the original image and reconstruct segmentation masks based on these features. While these models excel in tasks like polyp segmentation, their computational intensity poses a significant challenge, particularly for deployment on resource-constrained devices such as IoT platforms. To address this limitation, the proposed AGResU-NetV2 emerges as a promising solution. By leveraging attention mechanisms and integrating them with the lightweight backbone of MobileNetV2, AGResU-NetV2 strives to strike a balance between computational efficiency and segmentation performance.

4.2 Future Work

In future work, it would be beneficial to evaluate the performance of the proposed ensemble model on various datasets to compare its effectiveness against other existing models. Given the time constraints and the scope of this report, a potential avenue for further exploration involves implementing AGResU-NetV2 in Python using the Keras library. Additionally, integrating real IoT hardware prototypes for further testing could provide valuable insights into the model's capabilities in practical scenarios. Addressing the limitation of training data remains a significant challenge for deep learning models. Therefore, a key focus of future research will involve developing a clinical polyp segmentation model to bridge this gap and enhance the applicability of the proposed approach in real-world settings.

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Video link https://www.youtube.com/watch?v=y2opGVay5m4